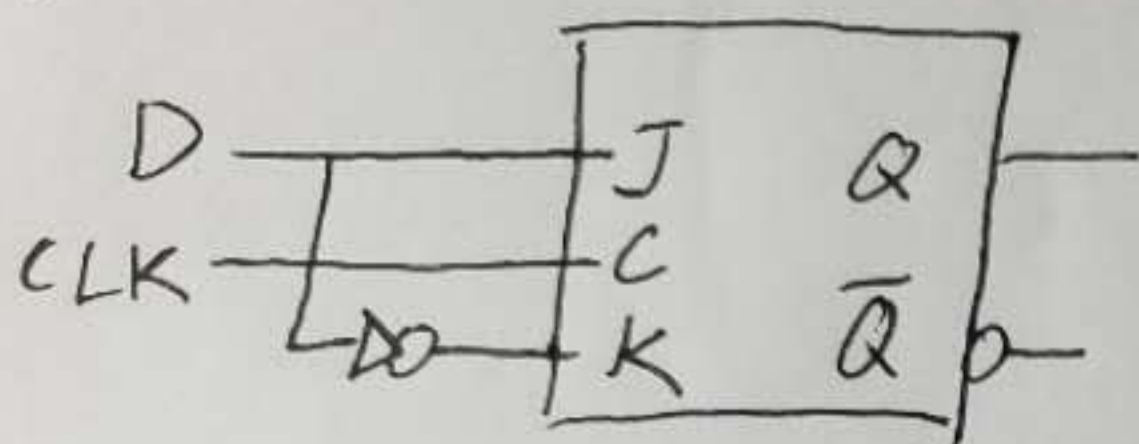


assignment 5

1.



2.

(1)

$$10 + 0 < 13 \text{ ns} \text{ 不能}$$

$$13 - 10 = 3 \text{ ns} \text{ 延时最少 } 3 \text{ ns}$$

$$T_{clk} > T_p + T_{pd} + T_{su} = 33 \text{ ns}$$

$$f_{max} = 1 / T_{clk} = 3.03 \times 10^7 \text{ Hz}$$

(2)

$$T_{clk} > 100 + 10 + 20 = 130 \text{ ns}$$

$$f_{max} = 1 / T_{clk} = 7.69 \times 10^6 \text{ Hz}$$

3.

略

能实现功能即可,对门电路数量无要求.